

Student Support Analytics

Business assessment of a student dropout prediction model

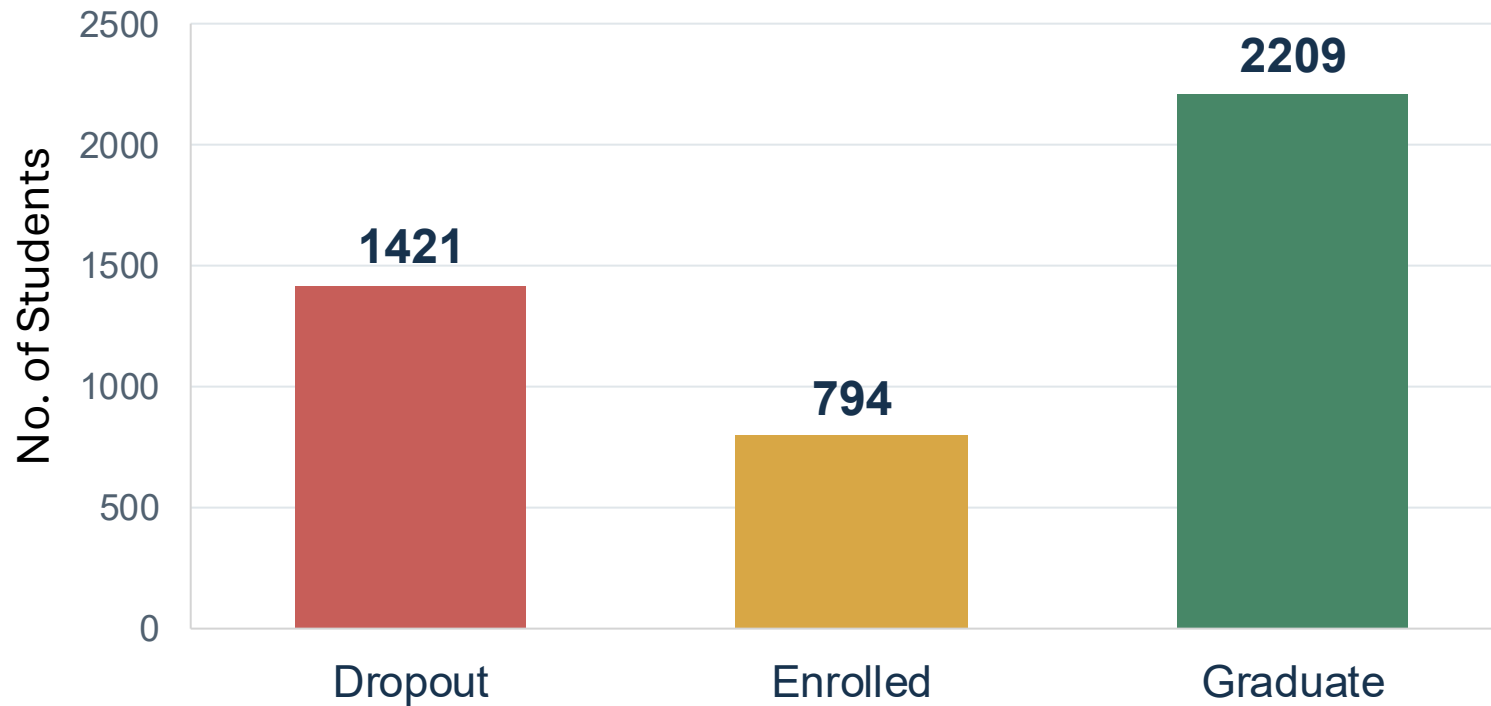
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This study evaluates a model that classifies students as Dropout, Enrolled, or Graduate.

Nearly One-Third of the Study Cohort Dropped Out



4,424

students

36 original predictors

Academic, demographic,
financial and economic
information.

32.1% dropped out in this dataset. This supports investigating more focused student assistance.

Predictions Could Help Staff Prioritize Student Support

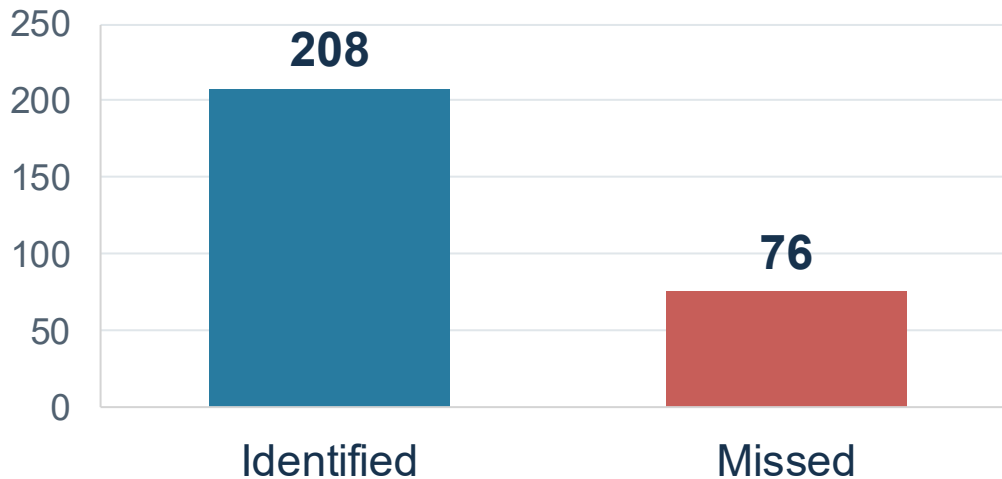
Stage	Proposed activity
1. Review information	Validate records available at an agreed prediction checkpoint.
2. Generate predictions	Identify cases for review, with uncertainty acknowledged.
3. Human review	Check records and consider circumstances absent from the model.
4. Offer assistance	Provide academic guidance or financial-support referrals.
5. Monitor outcomes	Record engagement, assistance received and subsequent outcomes.

Timing matters

The current model uses second-semester information. Prospective use requires predicting an outcome that has not already occurred. Enrollment-time prediction needs a separately evaluated model.

The Model Identified 73% of Actual Dropouts

Among actual dropouts

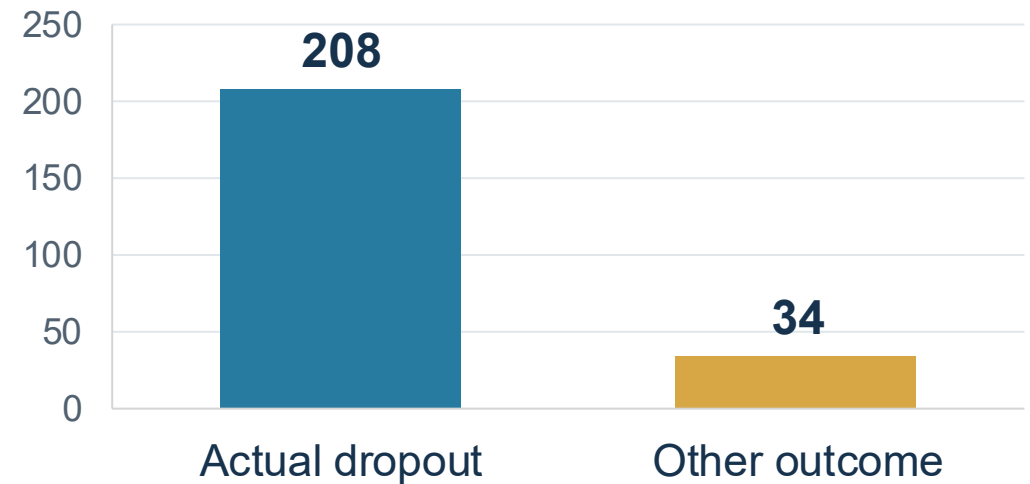


73.2% recall

The model identified approximately 73 of every 100 actual dropouts.

Overall accuracy: 76.2% (674 of 885).

Among dropout predictions



86.0% precision

Approximately 86 of every 100 dropout predictions were correct.

Institutional Value Depends on Improving Student Outcomes

A prediction creates value only when it supports an effective response that improves on existing practice

Potential benefit	Proposed measure
More focused outreach	Staff time, students contacted and completed reviews
Better access to assistance	Support uptake and unmet need across student groups
Improved retention	Additional students retained compared with usual practice
More efficient allocation	Program cost per additional student retained
Better student experience	Student feedback, complaints and correction requests

The project demonstrates predictive performance. It has not measured improvements in retention, staff productivity, or student satisfaction caused by model-assisted support.

ROI Requires Measured Costs and Additional Retention

Financial return cannot be established until program costs and retention improvements attributable to the intervention are measured.

Net benefit = additional retentions × net value per student – program cost

ROI = net benefit ÷ cost | Break-even retentions = cost ÷ net value per student

Required input	Evidence needed
Program cost	Integration, monitoring, staff review, outreach and assistance
Net value per retention	Finance estimate after incremental costs, for a defined period
Additional retention	Improvement attributable to support versus usual practice

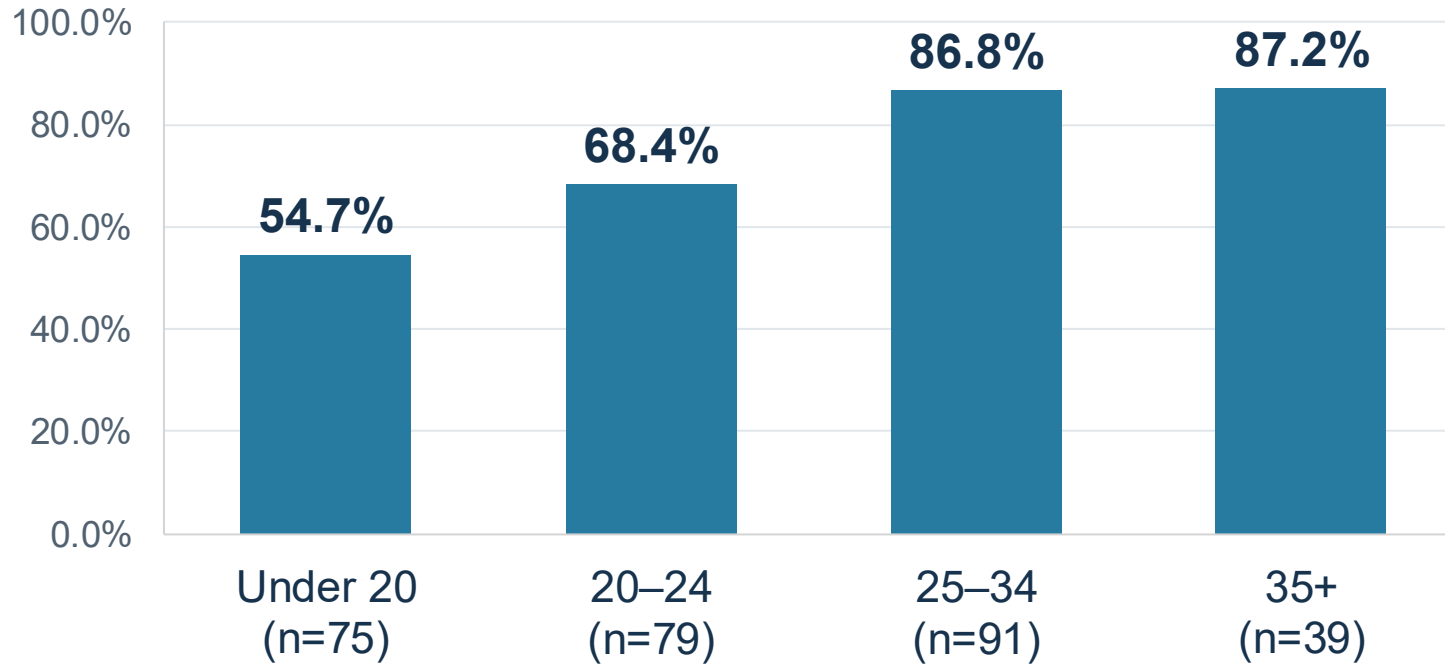
Human Review and Safeguards Are Essential

Students need a route to correct inaccurate information. Named institutional owners must oversee data, interventions, and evaluation.

Risk	Proposed safeguard
Students needing help are missed	Preserve staff referrals, self-referrals and standard support.
Incorrect labels cause stigma	Require human review and neutral, supportive communication.
Financial hardship shapes decisions	Offer assistance; prohibit punitive applications.
Inputs reveal an existing outcome	Verify timestamps and a prospective prediction checkpoint.
Sensitive records are exposed	Limit access, minimize stored data and define retention rules.
Performance changes over time	Monitor later cohorts, with a process to pause or revise use.

Younger dropouts were less often identified

Percentage of actual dropouts correctly identified, by age group.



34 of 75

actual dropouts
under 20 were **missed**.

An outreach process
based only on model flags
could **underserve**
younger students.

Among students with outstanding fees, seven of 15 non-dropouts were incorrectly labelled Dropout. This small denominator makes the estimated error rate uncertain (require investigation)

Broader Implementation Requires Prospective Evidence

A phased evaluation should establish local applicability before predictions influence student-support decisions.

Phase	Activities	Progression requirement
1. Readiness	Check local data, timestamps, intended use, ownership and privacy.	Inputs predict a genuinely future outcome.
2. Shadow evaluation	Generate predictions without changing services; observe later outcomes.	Review local performance and group errors against agreed criteria.
3. Controlled pilot	Compare additional supportive outreach with usual practice.	Assess incremental retention, costs, access and unintended effects.

The academic calendar determines the timeline. A short technical test cannot establish retention impact.

Recommendation: A Controlled Pilot

Proceed to readiness assessment and prospective evaluation before broader institutional use.

73.2%

of actual dropouts identified

86.0%

of dropout predictions correct

Conditions for broader use

- Verified timing and performance on future cohorts
- Evidence that support improves outcomes
- Acceptable cost, workload and group outcomes
- Human oversight and correction processes

Conclusion:

The project supports further evaluation of analytics-assisted student support. It does not yet establish financial returns, intervention effectiveness, or readiness for autonomous decisions.

References and evidence sources

DATASET

Realinho, V., Vieira Martins, M., Machado, J., & Baptista, L. (2021).
Predict Students' Dropout and Academic Success. UCI Machine Learning Repository.
<https://doi.org/10.24432/C5MC89>